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CHENGALPATTU DISTRICT

Class : 12 19.12.2025, FRIDAY

COMMON HALF YEARLY EXAMINATION, 2025 - 26

PHYSICS

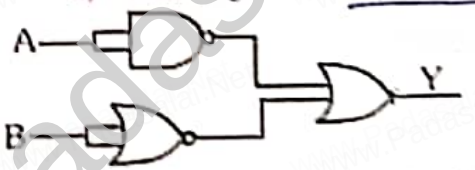
Time Allowed : 3.00 Hours] [Max. Marks : 70

PART-I

15x1=15

I. **Choose the correct answer.**

1. Atomic number of H-like atom with ionization potential 122.4 V for n = 1 is
   a) 1 b) 2 c) 3 d) 4
2. A carbon resistor of (47 ± 4.7) K Ω to be marked with rings of different colours for its identification. The colour code sequence will be
   a) Yellow - Green - Violet - Gold b) Yellow - Violet - Orange - Silver
   c) Violet - Yellow - Orange - Silver d) Green - Orange - Violet - Gold
3. The colours on the soap bubble is due to
   a) interference b) polarisation c) diffraction d) refraction
4. Consider an oscillator which has a charged particle oscillating about its mean position with a frequency of 300 MHZ. The wavelength of electromagnetic waves produced by this oscillator is
   a) 1 m b) 10 m c) 100 m d) 1000 m
5. The technology used for stopping the brain from processing pain is
   a) Precision medicine b) Wireless brain sensor
   c) Virtual reality d) Radiology
6. The maximum power is dissipated for an alternating current in
   a) LC circuit b) a pure resistive circuit
   c) a pure inductive circuit d) a pure capacitive circuit
7. A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. The length of the image is
   a) 2.5 cm b) 5 cm c) 10 cm d) 15 cm
8. The transverse nature of light is shown in
   a) interference b) diffraction c) scattering d) polarisation
9. The output Y of the given logic gate is similar to the output of a
   a) OR gate b) NOR gate
   c) AND gate d) NAND gate



10. An electric field $\vec{E} = 10x\hat{i}$ exists in a certain region of space. Then the potential difference $V = V_0 - V_A$, where V_0 is the potential at the origin and V_A is the potential at $x = 2m$ is
    a) 10 V b) -20 V c) +20 V d) -10 V
11. A circular coil with a cross-sectional area of 4 cm² has 10 turns. It is placed at the centre of a long solenoid that has 15 turns / cm and a cross sectional area of 10 cm². The axis of the coil coincides with the axis of a solenoid. What is their mutual inductance?
    a) 7.54 μH b) 8.54 μH c) 9.54 μH d) 10.54 μH
12. The radius of $_{30}Zn^{64}$ nucleus is nearly (in fermi)
    a) 1.2 b) 2.4 c) 3.6 d) 4.8
13. Two radiations with photon energies 0.9 eV and 3.3 eV respectively are falling on a metallic surface successively. If the work function of the metal is 0.6 eV, then the ratio of maximum speeds of emitted electrons in the two cases will be
    a) 1 : 4 b) 1 : 3 c) 1 : 1 d) 1 : 9
14. To obtain sustained oscillation in an oscillator,
    a) feed back should be positive b) feed back factor must be unity
    c) phase shift must be 0 or 2π d) all the above
15. SI unit of magnetic moment is
    a) Am^2 b) Am c) NAm^{-2} d) $NA^{-1}m^{-1}$

PART - II

II. **Answer Any SIX of The Following. (Answer Question No.24 Compulsory)** 6x2 =12

16. What is meant by excitation energy?

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17. Define temperature coefficient of resistance.
18. Two light sources have intensity of light as I_0. What is the resultant intensity at a point where the two light waves have a phase difference of $\pi/3$?
19. State Snell's law or law of refraction.
20. Give two uses of infrared (IR) radiation.
21. Define electric flux.
22. A straight metal wire crosses a magnetic field of flux 4 m Wb in a time of 0 4s. Find the magnitude of the emf induced in the wire.
23. Is an ammeter connected in series or parallel in a circuit? Why?
24. Prove the following Boolean expressions using the laws and theorems of Boolean algebra
    (i) $A(\overline{A}+B) = AB$ (ii) $(A + B)(A + C) = A + BC$

PART - III

III. Answer Any SIX of The Following. (Answer Question No.33 Compulsory) 6x3=18

25. Write down any three drawbacks or limitations of Bohr atom model.
26. Explain the equivalent resistance of a parallel resistor network.
27. A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6} T, then calculate the current which gives a deflection of 60°.
28. Differentiate polarised and unpolarised light.
29. List out the advantages and limitations of frequency modulation.
30. An object of 4 cm height is placed at 6 cm in front of a concave mirror of radius of curvature 24 cm. Find the position, height and magnification of the image.
31. Assuming that the length of the solenoid is large when compared to its diameter, find the equation for its inductance.
32. Derive the expression for de Broglie wavelength of electrons.
33. A parallel plate capacitor filled with mica having Er= 5 is connected to a 10V battery. The area of each parallel plate is 6 cm^2 and separation distance is 6 mm. Find the capacitance and stored charge.

PART - IV

IV. Answer All Questions. 5x5 =25

34. a) Obtain a relation for the magnetic field at a point along the axis of a circular coil carrying current using Biot Savart law.

(OR)

b) Obtain Einstein's photoelectric equation with necessary explanation.

35. a) Explain in detail the construction and working of a Vande Graff generator.

(OR)

b) (i) What is half - life of a radioactive nucleus? Give the expression.
(ii) Calculate the number of nuclei of carbon-14 undecayed after 22, 920 years if the initial number of carbon-14 atoms is 10,000. The half life of carbon-14 is 5730 years.

36. a) Draw the circuit diagram of a half wave rectifier and explain its working.

(OR)

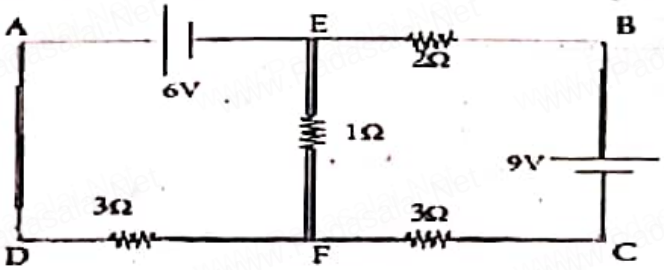
b) Explain the Maxwell's modification of Ampere's circuital law.

37. a) Explain the construction and working of a transformer.

(OR)

b) Obtain lens maker's formula.

38. a) (i) State Kirchoff's voltage rule.
(ii) Calculate the current that flows in the 1 Ω resistor in the following circuit.



(OR)

b) Obtain the equation for bandwidth in Young's double slit experiment.

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